MIL-P-12815B (AR)
8 February 1980
SUPERSEDING
MIL-P-12815A (ORD)
23 August 1960

MILITARY SPECIFICATION

POWER SOURCE PIEZOID (LUCKY)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 The specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing for flat type polarized polycrystalline ceramic piezo-electric elements designated as Power Source, Piezoid (Lucky).

2. APPLICATION DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1315

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-1167 - Ammunition Data Card
MIL-STD-1169 - Packaging, Packing and Marking For Shipment of Inert Ammunition Components

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Power source, piezoid (lucky). The lucky shall comply with all requirements specified on the applicable drawings and specifications.

3.3 Marking of polarity. The lucky may be marked on the positive electrode so as to identify the direction of polarization. The mark shall be a black dot and shall be of such contrast to the background that it is readily discernible, and shall not extend beyond the silvered area and shall occupy not less than three nor more than nine percent of the silvered surface area, unless otherwise specified on the applicable drawings.

3.4 Activity. The lucky shall have a piezoelectric coefficient, d_{33}, of not less than 9×10^{-10} coulombs per pound where d_{33} relates the charge to the load and is measured along the axis of polarization. The piezoelectric coefficient is determined by measuring the charge developed while releasing the lucky from a compressive pressure 1000 pounds per square inch (psi) for units of 0.012 to .500 square inch electrode area and 500 psi for .501 and 1.000 square inch areas.

3.5 Dielectric constant. The lucky shall exhibit a dielectric constant of not less than 2000 relative to air.

3.6 First article inspection/test. This specification contains technical provisions for a First Article Inspection/Test. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

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3.7 Workmanship. The lucky shall be free of dirt, grease and other foreign material. The cleaning method shall not be injurious to the power source nor shall it be contaminated by the cleaning agent used. Silver electrode shall not extend onto edge of lucky.

3.8 Process description and controls. The contractor in accordance with the contract provisions shall make available a written description of the specified production process and process controls, determined by the Government as crucial for selected components that are to be used in production of the first article sample (see par 4.3.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 25 luckies produced and packed using the same production processes, procedures and equipment which will be used in fulfilling the contract. All materials, packaging and packing shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. See MIL-A-48078. All First Article Samples shall be subjected to all inspections/tests specified in paragraph 4.4.2.1 herein.

4.3.3 Rejection. See MIL-A-48078.

4.3.4 Process controls. At the same time as submission of the First Article Sample, the contractor shall make available a written description of the production process and controls being used.

4.4 Quality conformance inspection.

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4.4.1 Lot formation. A lot shall consist of luckies produced by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting performance, safety, or fit as determined by the Government, shall not be cause for changing the interfix number.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Lucky				See Applicable Dwg.
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Polarity (if applicable)		100%	3.3	Visual
102	Activity		100%	3.4	Test/4.5.1
103	Dielectric constant		100%	3.5	Test/4.5.2
104	Outside diameter, max.		0.40%	3.2	Gage
105	Diameter of silver coating		0.40%	3.2	Gage
106	Thickness		0.40%	3.2	Gage
107	Sides of lucky not parallel		0.40%	3.2	Gage
108	Sides of lucky not flat		0.40%	3.2	Gage
109	Thickness of silver coating		0.40%	3.2	Gage
110	Lead slot missing (if applicable)		0.40%	3.2	Visual
111	Resistance of silver coating		0.40%	3.2	Test/See applicable dwg.
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual
202	Slot contour (if applicable)		0.65%	3.2	Gage

NOTES:

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4.4.3 Testing

4.4.3.1 Activity and dielectric constant. These tests shall be performed 100%. In the event that the inspector discovers that the test has not been fully complied with or that non-conforming units have not been rejected, he shall reject all units tested since his last satisfactory surveillance. The contractor shall retest these rejected units 100 percent in the presence of the inspector. Units failing to meet the requirements during retest shall be classified defective and removed from the lot.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Activity. After polarizing, the lucky shall be heated to a temperature of 165 degrees Fahrenheit (^{o}F) $\pm$ 10°F and maintained at that temperature for a period of 10 minutes. They shall then be cooled to 75° $\pm$ 10°F and maintained at that temperature for not less than 24 hours and shall then be tested to determine conformance to this specification.

4.5.2 Determination of dielectric constant. After performing the test specified in 4.5.1 the dielectric constant shall be determined using the following formula:

$$K = \frac{4.44\ C\ t}{A}$$

Where:

K = dielectric constant of lucky
*C = Capacitance of lucky in micro-microfarads
t = maximum thickness of lucky (inches)
A = minimum area of lucky electrode surface (square inch)

*Capacitance shall be determined using a bridge or meter which applies a measuring signal of 1,000 cycles per second. The test shall be conducted at a temperature between +65°F and +85°F.

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5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level B. The lucky shall be preserved, packaged and marked in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Intended use. This general specification shall be used for flat bype piezoid power source (lucky).

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Submission of ammunition data cards. In addition to the normal distribution of records when the lucky is procured by the Army Armament Materiel Readiness Command, one copy of all ammunition data cards shall be forwarded to the following: Commander, ARRADCOM, ATTN: DRDAR-QAR-E, Dover, NJ 07801; and Commander, ARRCOM, ATTN: DRSAR-QAD, Rock Island, IL 61299.

6.5 It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items for delivery may be equivalent to such contract quantities provided the lot formation criteria and sampling provisions of this specification are maintained.

6.6 Delay in distribution. Document was approved 8 February 1980, however, it was not received by Department of Defense Single Stock Point for indexing and distribution until February 1983.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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